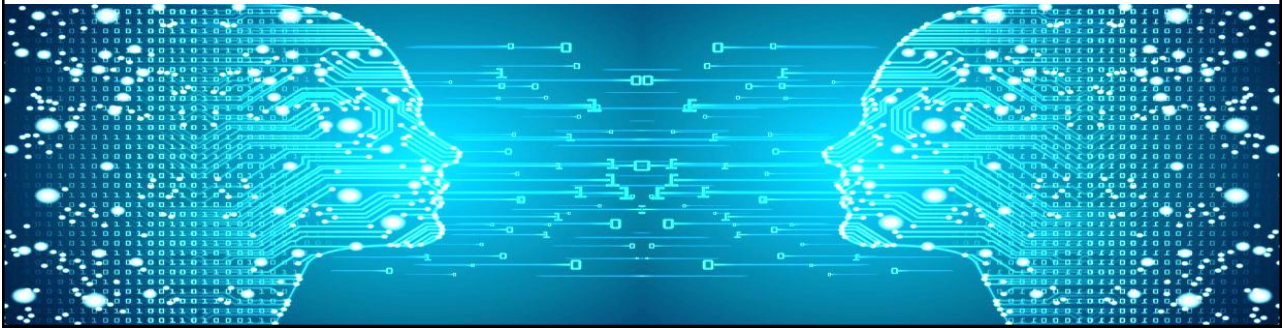
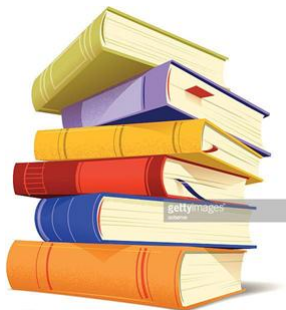


Machine Learning



k-Means Clustering



- Introduction
- Description of the Method
- The k-Means Clustering Algorithm

Introduction

- k-Means clustering is:
 - An unsupervised learning algorithm
 - To partition N observations into k clusters (k being an integer).

Description of the method

- Given N data points $x_1, x_2, \dots, x_N$, each x_j having m attributes. The aim is to group this data into K clusters $C_1, C_2, \dots, C_K$, centered at mean points $\mu_1, \mu_2, \dots, \mu_K$.
- **Two Core Conditions**
 1. **Exclusive Membership (Assignment Rule):** Each point x_i must belong to exactly one cluster C_j .
A point x_i is assigned to cluster C_j if and only if μ_j is the closest mean to x_i .
 2. **Minimizing Objective Function:** The choice of central points μ_k must **minimize** the functional J :

$$J(\mu_1, \dots, \mu_K) = \sum_{k=1}^K \sum_{x_j \in C_k} \|x_j - \mu_k\|^2$$

- J measures the sum of squared distances of points from their respective cluster means (within-cluster variation).

Description of the Method

▪ The method:

1. Initialize k cluster centers.
2. Assign each data point to the cluster with the nearest center.
3. Recompute each cluster center as the mean of the points in that cluster.
4. Repeat steps 2 and 3 until convergence (the centers no longer change or change only slightly).

- **Initialization sensitivity:** Each run of the k -Means algorithm may yield different means and clusters due to the random selection of K initial mean values.

The k -Means Clustering Algorithm

Step	Action
1. Initialization	Set $j = 0$. Choose K random initial means $\mu_1^{(0)}, \dots, \mu_K^{(0)}$.
2. Assignment (E-step)	For each point x_i , find the closest mean $\mu_k^{(j)}$ and assign x_i to that cluster C_k .
3. Update (M-step)	Set $j = j + 1$. Recalculate the new mean $\mu_k^{(j)}$ for all points within the updated cluster C_k .
4. Iteration Check	Repeat Steps 2 and 3 until convergence.
5. Final Output	Return the final cluster labels L , the optimized means μ , and the within-cluster sum of distances Sum .

Summary

- K-means is an unsupervised clustering algorithm that partitions data into k groups.
- The goal is to minimize the distance between points and their assigned cluster centers.
- Workflow: initialize k centroids, assign points to the nearest centroid, recompute centroids, and iterate until convergence.
- Strengths: simple, fast, efficient for large datasets.
- Limitations: requires predefined k , sensitive to initialization and outliers, works best with spherical clusters.

